MSG No. 122.335 $I\bar{4}'2'd$ [Type III, tetragonal]

Table 1: Wyckoff site: `4a`, site symmetry: `-4'..`

No.	position	mapping
1	$[0,\ 0,\ 0]$	`[1,2,7,8]`
2	$\left[0,\ \frac{1}{2},\ \frac{1}{4}\right]$	`[3,4,5,6]`
3	$\left[\frac{1}{2},\ \frac{1}{2},\ \frac{1}{2}\right]$	`[9,10,15,16]`
4	$\left[\frac{1}{2},\ 0,\ \frac{3}{4}\right]$	`[11,12,13,14]`

Table 2: Wyckoff site: `4b`, site symmetry: `-4'..`

No.	position	mapping
1	$\left[0,\ 0,\ \frac{1}{2}\right]$	`[1,2,7,8]`
2	$\left[0,\ \frac{1}{2},\ \frac{3}{4}\right]$	`[3,4,5,6]`
3	$\left[\frac{1}{2},\ \frac{1}{2},\ 0\right]$	`[9,10,15,16]`
4	$\left[\frac{1}{2},\ 0,\ \frac{1}{4}\right]$	`[11,12,13,14]`

Table 3: Wyckoff site: `8c`, site symmetry: `2..`

No.	position	mapping
1	$[0,\ 0,\ z]$	`[1,2]`
2	$\left[0,\ \frac{1}{2},\ z+\frac{1}{4}\right]$	`[3,4]`
3	$\left[0,\ \frac{1}{2},\ \frac{1}{4}-z\right]$	`[5,6]`
4	$[0,\ 0,\ -z]$	`[7,8]`
5	$\left[\frac{1}{2},\ \frac{1}{2},\ z+\frac{1}{2}\right]$	`[9,10]`
6	$\left[\frac{1}{2},\ 0,\ z+\frac{3}{4}\right]$	`[11,12]`
7	$\left[\frac{1}{2},\ 0,\ \frac{3}{4}-z\right]$	`[13,14]`
8	$\left[\frac{1}{2},\ \frac{1}{2},\ \frac{1}{2}-z\right]$	`[15,16]`

Table 4: Wyckoff site: `8d`, site symmetry: `.2'.`

No.	position	mapping
1	$\left[x,\ \frac{1}{4},\ \frac{1}{8}\right]$	`[1,5]`
2	$\left[-x,\ \frac{3}{4},\ \frac{1}{8}\right]$	`[2,6]`
3	$\left[\frac{3}{4},\ \frac{1}{2}-x,\ \frac{3}{8}\right]$	`[3,15]`
4	$\left[\frac{1}{4},\ x+\frac{1}{2},\ \frac{3}{8}\right]$	`[4,16]`
5	$\left[\frac{1}{4},\ -x,\ \frac{7}{8}\right]$	`[7,11]`
6	$\left[\frac{3}{4},\ x,\ \frac{7}{8}\right]$	`[8,12]`
7	$\left[x+\frac{1}{2},\ \frac{3}{4},\ \frac{5}{8}\right]$	`[9,13]`
8	$\left[\frac{1}{2}-x,\ \frac{1}{4},\ \frac{5}{8}\right]$	`[10,14]`

Table 5: Wyckoff site: 16e, site symmetry: 1

No.	position	mapping
1	$[x,\, y,\, z]$	[1]
2	$[-x,\, -y,\, z]$	[2]
3	$\left[-y,\, \frac{1}{2}-x,\, z+\frac{1}{4}\right]$	[3]
4	$\left[y,\, x+\frac{1}{2},\, z+\frac{1}{4}\right]$	[4]
5	$\left[x,\, \frac{1}{2}-y,\, \frac{1}{4}-z\right]$	[5]
6	$\left[-x,\, y+\frac{1}{2},\, \frac{1}{4}-z\right]$	[6]
7	$[y,\, -x,\, -z]$	[7]
8	$[-y,\, x,\, -z]$	[8]
9	$\left[x+\frac{1}{2},\, y+\frac{1}{2},\, z+\frac{1}{2}\right]$	[9]
10	$\left[\frac{1}{2}-x,\, \frac{1}{2}-y,\, z+\frac{1}{2}\right]$	[10]
11	$\left[\frac{1}{2}-y,\, -x,\, z+\frac{3}{4}\right]$	[11]
12	$\left[y+\frac{1}{2},\, x,\, z+\frac{3}{4}\right]$	[12]
13	$\left[x+\frac{1}{2},\, -y,\, \frac{3}{4}-z\right]$	[13]
14	$\left[\frac{1}{2}-x,\, y,\, \frac{3}{4}-z\right]$	[14]
15	$\left[y+\frac{1}{2},\, \frac{1}{2}-x,\, \frac{1}{2}-z\right]$	[15]
16	$\left[\frac{1}{2}-y,\, x+\frac{1}{2},\, \frac{1}{2}-z\right]$	[16]